GEOG 172

Spatial Sampling Schemes

Lecture 3-2

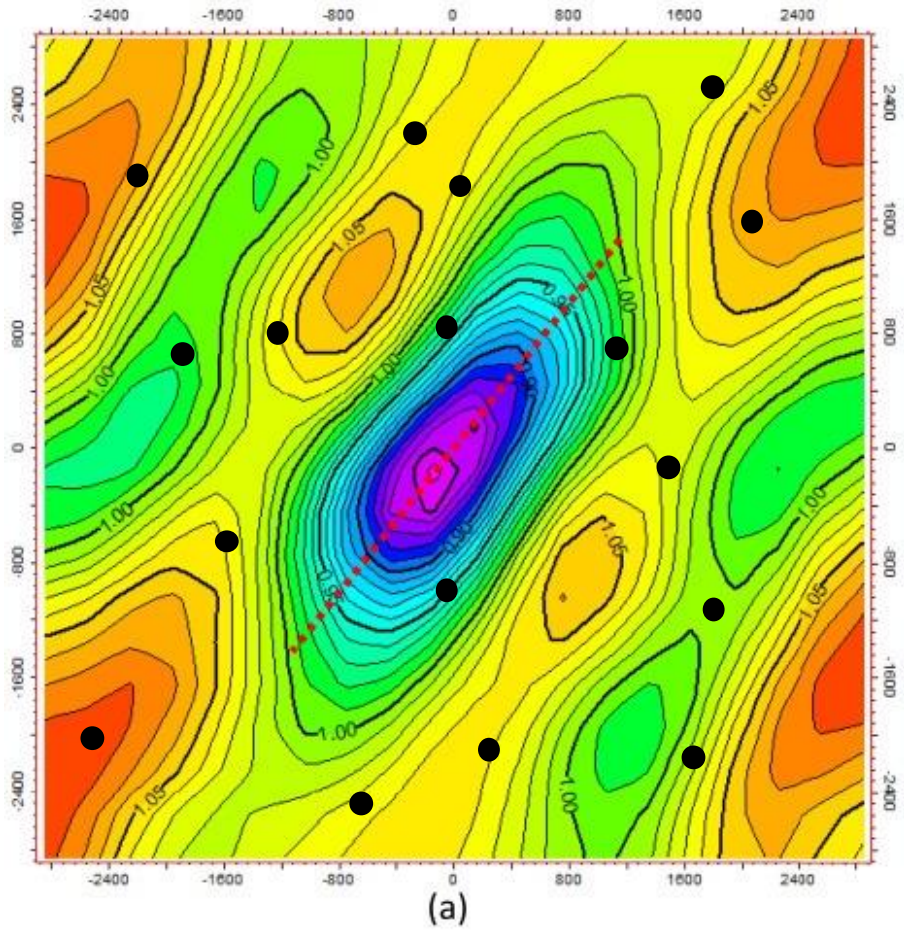
SPATIAL SAMPLING

The process of selecting a subset of elements from a population, and collecting observations of some attribute(s) of those individuals that can then be used to estimate characteristics of the entire population

WHY SAMPLE AT ALL?

Spatial samples are used to make inferences about unobserved populations of processes. They are desirable because

1. Population size, or the size of region of interest, make exhaustive measurement impractical
2. Some geographic locations are inaccessible (sometimes or all the time)
3. At times only a certain level pf precision is needed to reduce the effect of errors
4. The estimation efficiency of a sample is sensitive to the spatial structure (1st and 2nd order effects)
5. Problematic datasets require additional data collection



SPATIAL COVERAGE

This sample has good spatial coverage

FEATURE COVERAGE

This sample has poor feature coverage

Spatial Sampling Schemes

A story in pictures, then in a computer

PROBABILITY DESIGNS

Selection is governed by some form of random mechanism

1. Simple Random Sampling
2. Systematic Random Sampling
3. Stratified Random Sampling
4. Clustered Sampling
5. Hierarchical Sampling

NON-PROBABILITY DESIGNS

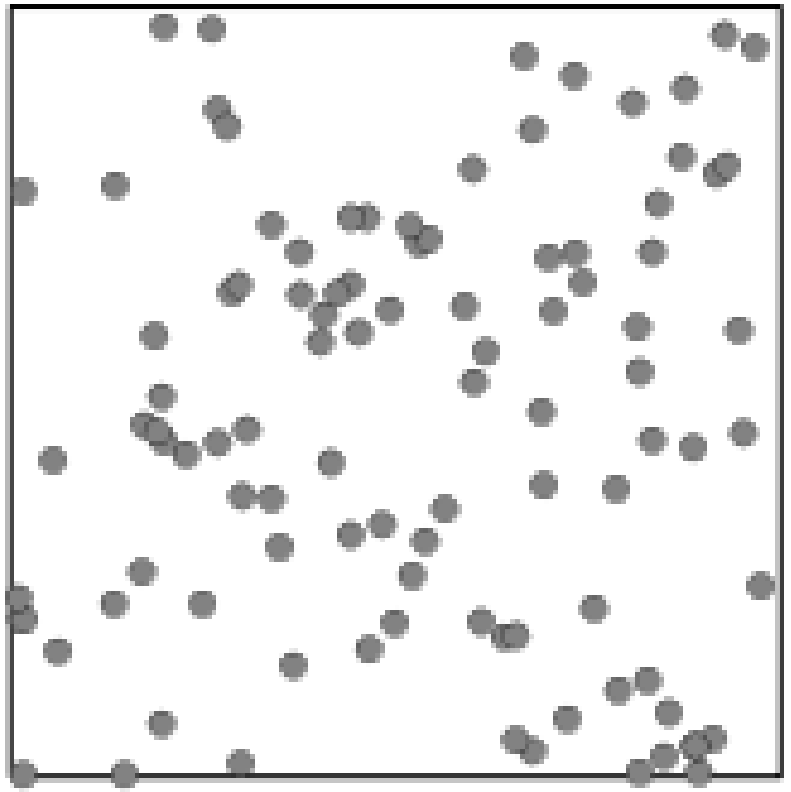
Selection does not rely on a random mechanism

1. Purposive Sampling
2. Quota Sampling
3. Snowball Sampling

SIMPLE RANDOM

Choosing randomly a set of m sample points in A, where each location has an equal probability of being sampled.

While simple to implement, this design may not produce a distribution of sample locations that covers A or the underlying spatial structure of the z(s)



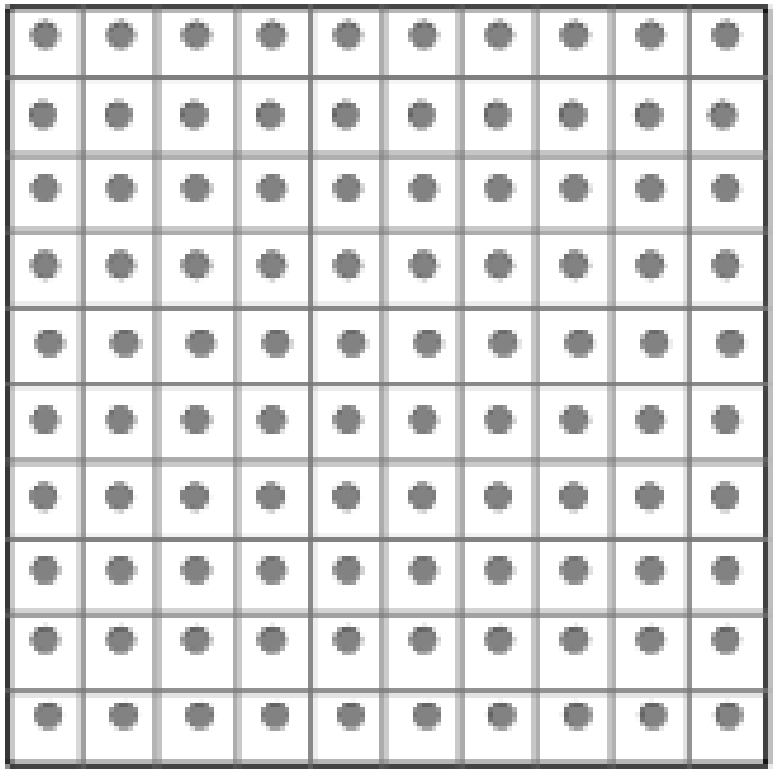
(a) random

SYSTEMATIC RANDOM

The study area is segmented into equal intervals and sample points are drawn along those intervals following a specified rule

Sub-regional configuration may take a variety of shapes, but square, triangular, and hexagonal are the most common

Systematic sampling schemes assure the spread of observation locations across a study area and provide better spatial coverage than random sampling



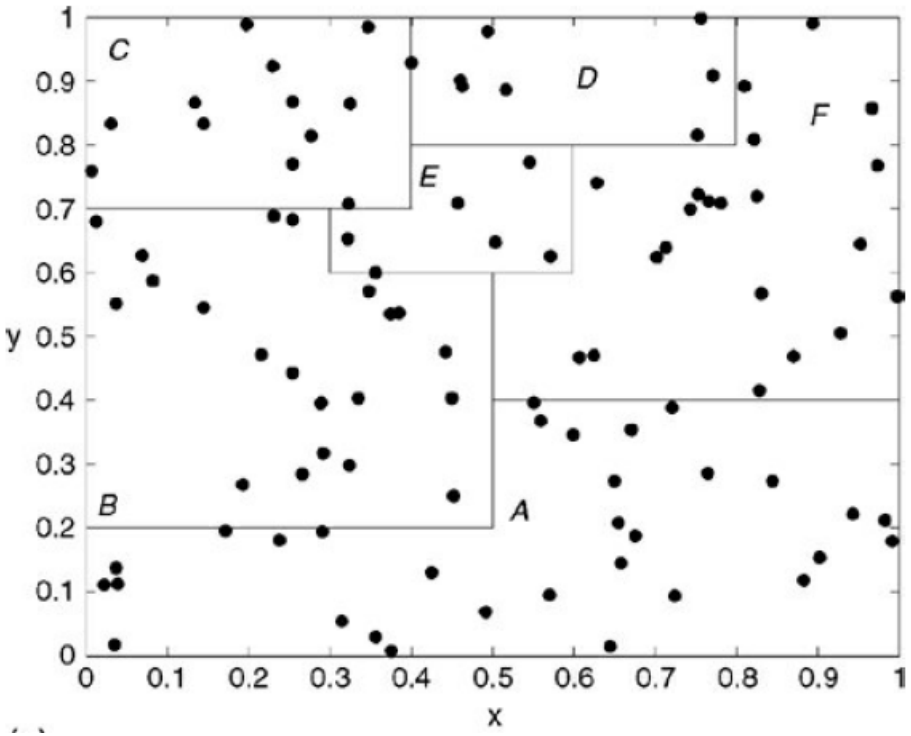
(c) systematic centric

STRATIFIED RANDOM

A is first partitioned into non-overlapping strata and a set of samples is collected in each strata. The sum of all sample points across all strata equals *m*

Consideration of the underlying process and spatial non-stationarity are important for determining the size, shape, and location of strata

Ideally, strata will align with the extent of first-order variation creating stationary sub-regions

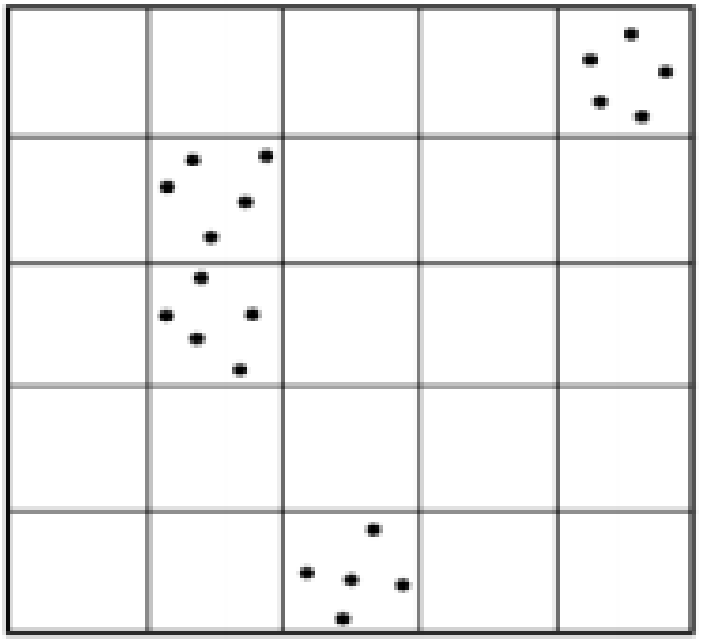


CLUSTERED

First, the population is grouped into clusters from which a sub-set of clusters is randomly selected. For those selected clusters, individuals are then sampled, usually randomly

Used when data present natural grouping and to save travel costs

Case 5:
Random Point Sample within Clusters
(two-stage cluster sample)



HIERARCHICAL

First, a level-1 partitioning of *A* is created and a subset of those sub-regions is selected. For those sub-regions, a level-2 partitioning is created and a second round of random selection is completed. This process continues until the researcher reaches their scale of interest

- Useful when scale dependencies are of primary interest
- Related schemes that control the distance between points have been developed to address the masking of scale effects by variation in inter-sample distances

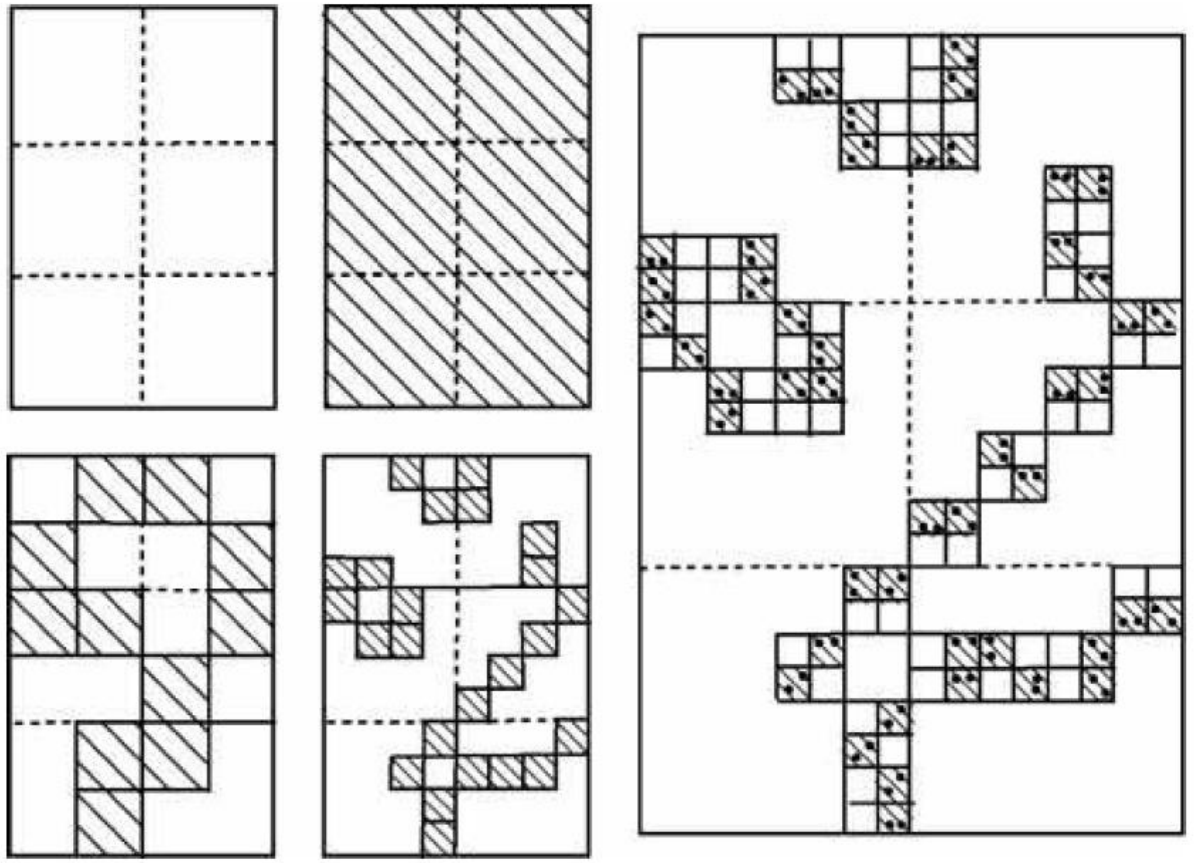
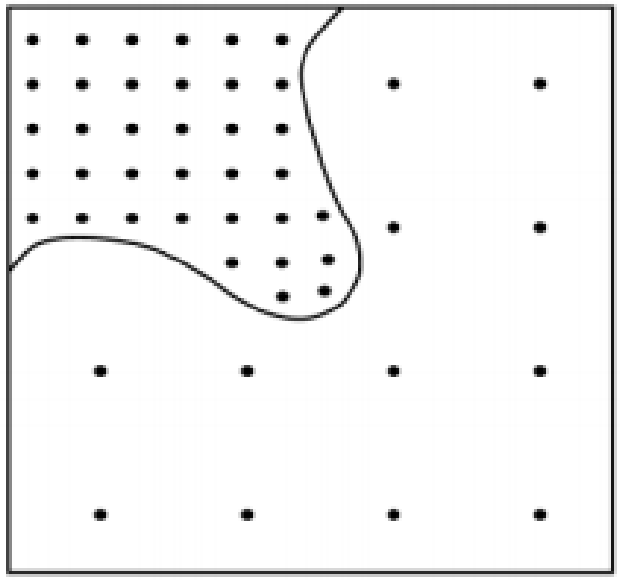
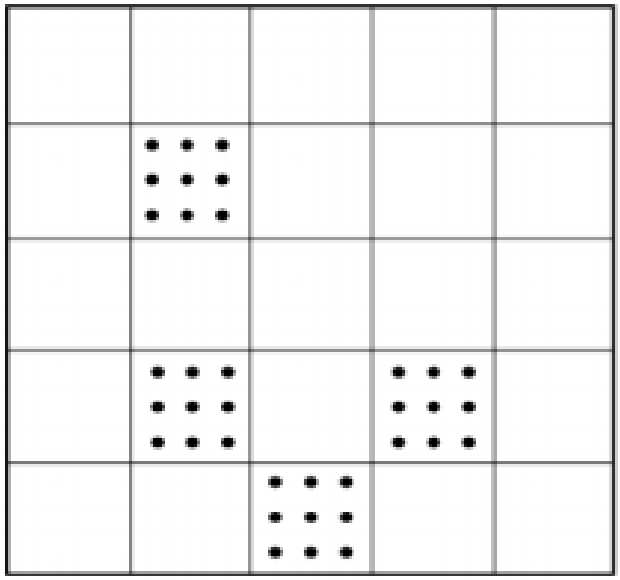


Figure 3.3 A nested sampling scheme

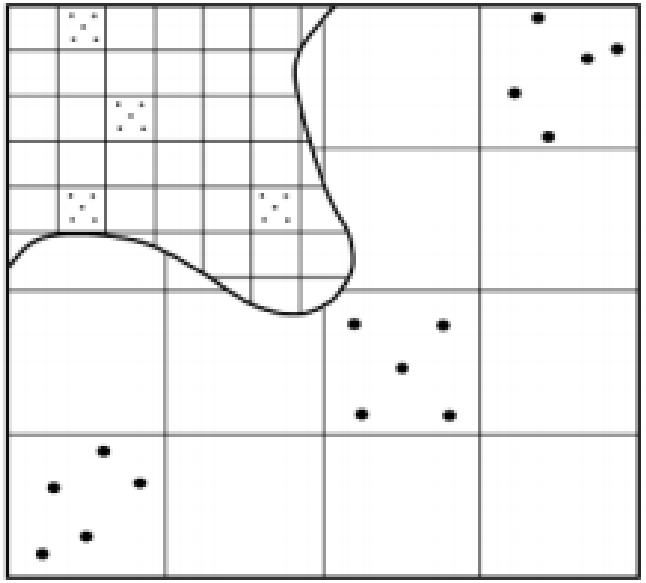
Case 2: Disproportional Stratified Systematic Aligned



Case 3: Cluster Systematic



Case 4: Disproportional Stratified Cluster



NON-PROBABILITY DESIGNS

Although generally not appropriate for statistical inference, non-probability sampling does have an important place in quantitative reasoning and the formation of persuasive arguments

Purposive Sampling. samples are collected with a clear purpose in mind

Quota Sampling. Sampling proceeds until a minimum is reached for subsets of the population defined by some additional attribute

Snowball Sampling. After sampling an initial individual, sampling continues following the recommendations or relationships of that individual with others in the population

Building Sampling Schemes

From concept to computer

TUTORIAL

Does the sampling scheme you choose affect the inferences you make when different spatial processes are responsible for the spatial pattern you only partially observe?

Structure:

1. Introduction (Thur *Right Now*)
2. Spatial Data Generating Processes (Tue Demo)
3. Monte Carlo Simulation (Tutorial with TAs)
4. Spatial Sampling Schemes (Thur Demo)
5. Sampling Experiment (Tutorial -> Take Home Work)

FORMALLY

A sampling strategy consists of

1. **Sample Design.** The rule for choosing which locations to observe, including the probability that each location is selected (e.g., simple random sample)
2. **Estimator.** The statistical procedure applied to the observed sample values to estimate an unknown population quantity (e.g., sample mean)

FORMALLY

Two theories of sampling and inference

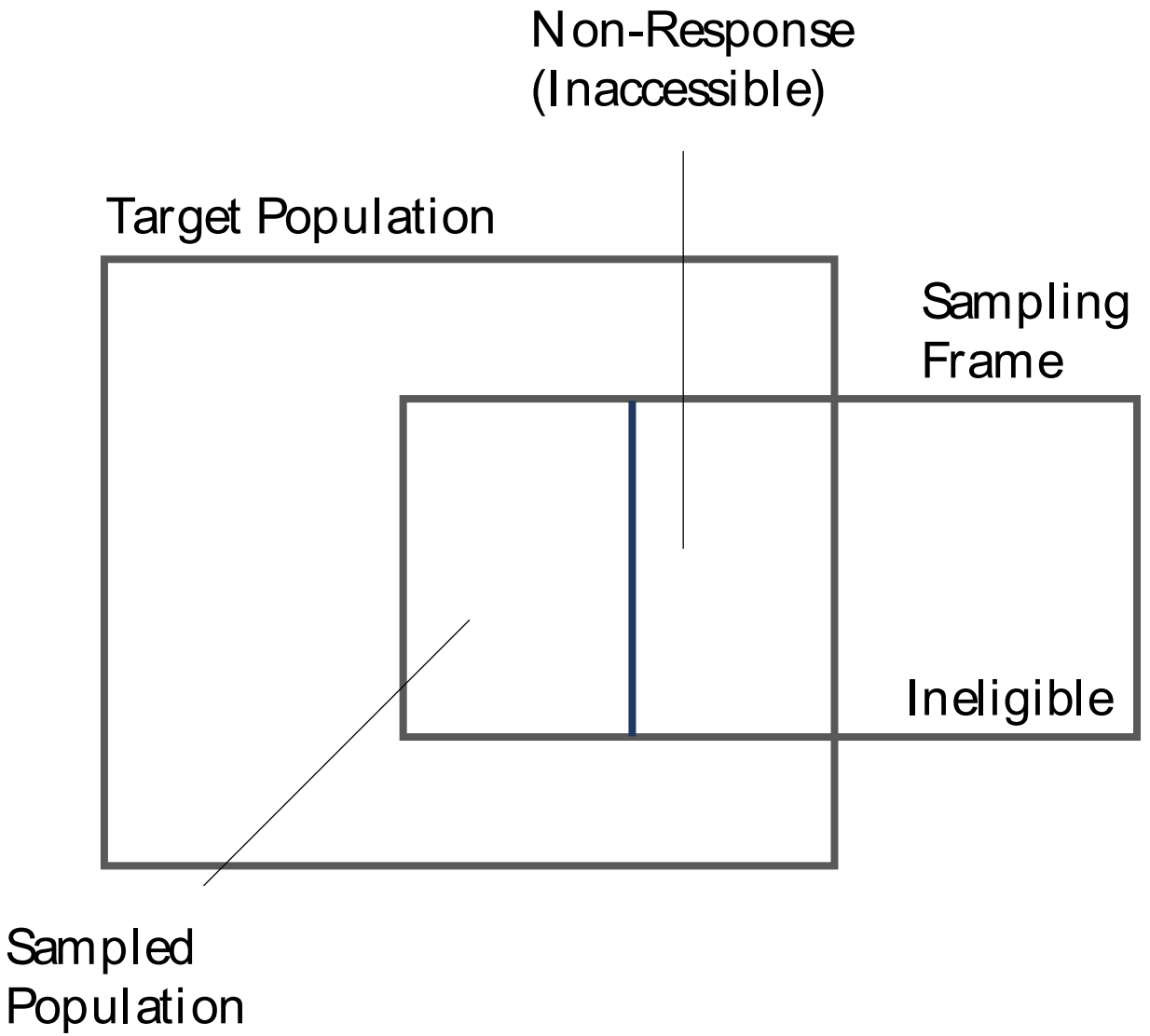
1. **Design-based.** An approach in which the values across the study area are treated as fixed and unknown, and all randomness comes from the random selection of sample locations.
2. **Model-based.** An approach in which the values across the study area are treated as one realization of a random field, and randomness comes from the process that generated the observed surface.

Random Field. A model of the variable being studied in which the value at every location in the study area is treated as a random quantity

FORMALLY

Key concepts in sampling are

1. **Target Population**. The set of elements (or locations) of interest to the problem under study.
2. **Sampling Frame**. The set of locations that can actually be observed, from which sample points are chosen.
3. **Sample.** The subset of locations selected from the sampling frame, together with the values of the variable observed at those locations.



STEPS INVOLVED IN SAMPLING

1. **Clarify the sampling objectives**. Is the objective about a 'here and now population' or about the superpopulation that generates the observed cross-sectional population?
2. **Define the population** to be sampled and the sampling units.
3. **Chose the sampling method** and compute the sample size under specific budget and precision constraints (e.g., a field survey, aerial survey, satellite-based survey)
4. **Design of a sampling plan** that describes where, when and how sampling units are to be taken.
5. **Draw the sample** and take measurements on the sampling units obtained.
6. **Analyze of the sample** in light of prior knowledge to obtain the features of the stochastic field, such as randomness, trend, autocorrelation, or stratification.